

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022**(CBCSS-UG)****Mathematics****MTS 6B 14 (E01)—GRAPH THEORY****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 14 (E01)---GRAPH THEORY

(Multiple Choice Questions for SDE Candidates)

1. What is the maximum number of edges in a bipartite graph having 12 vertices ?
 - (A) 24.
 - (B) 38.
 - (C) 36.
 - (D) 32.
2. What is the regularity of K_n ?
 - (A) n .
 - (B) $n - 1$.
 - (C) 0.
 - (D) 1.
3. A connected planar graph having 6 vertices, 7 edges contains _____ regions.
 - (A) 1.
 - (B) 3.
 - (C) 5.
 - (D) 7.
4. For a given graph G having v vertices and e edges which is connected and has no cycles, which of the following statements is true ?
 - (A) $v = e$.
 - (B) $v = e + 1$.
 - (C) $v + 1 = e$.
 - (D) $v = e - 1$.
5. A connected undirected graph containing n vertices and $n - 1$ edges _____.
 - (A) Cannot have cycles.
 - (B) Must contain at least one cycle.
 - (C) Can contain atmost two cycles.
 - (D) Must contain atleast two cycles.
6. Which of the following statements is/are TRUE for Tree ?

P : Every two points of G are joined by a unique path.

Q : G is connected and $p = q - 1$.

 - (A) P only.
 - (B) Q only.
 - (C) Both P and Q.
 - (D) Neither P and Q.
7. Let G be a simple graph with every pair of vertices is connected. Then G is :
 - (A) Trivial.
 - (B) Complete.
 - (C) Disconnected.
 - (D) Self complementary.

8. Let $G = K_n$ where $n > 5$. Then number of edges of any induced sub graph of G with 5 vertices :
- (A) 10. (B) 5.
(C) 6. (D) 8.
9. Number of edges incident with the vertex V is called ?
- (A) Degree of a graph. (B) Handshaking lemma.
(C) Degree of a vertex. (D) None of the above.
10. A graph with n vertices will definitely have a parallel edge or self loop if the total number of edges are :
- (A) Greater than $n - 1$. (B) Less than $n(n - 1)$.
(C) Greater than $\frac{n(n-1)}{2}$. (D) Less than $\frac{n^2}{2}$.
11. If G is the forest with 54 vertices and 17 connected components, G has _____ total number of edges.
- (A) 37. (B) 71.
(C) 17. (D) 54.
12. How many of the following statements are correct ?
1. All cyclic graphs are complete graphs.
 2. All complete graphs are cyclic graphs.
 3. All paths are bipartite.
 4. All cyclic graphs are bipartite.
 5. There are cyclic graphs which are complete.
- (A) 1. (B) 2.
(C) 3. (D) 4.
13. Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of bounded faces in any embedding of G on the plane is equal to :
- (A) 4. (B) 5.
(C) 6. (D) 7.

14. Radius of a graph, denoted by $\text{rad}(G)$ is defined by _____.

(A) $\text{Max}\{e(v) : v \in V(G)\}.$

(B) $\text{Min}\{e(v) : v \in V(G)\}.$

(C) $\text{Max}\{d(v, v) : v, u \in V(G)\}.$

(D) $\text{Min}\{d(u, v) : v, u \in V(G)\}.$

15. If a graph is planar, then it is embeddable on a :

(A) Circle.

(B) Square.

(C) Sphere.

(D) Triangle.

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 14 (E01)—GRAPH THEORY

(2019 Admissions)

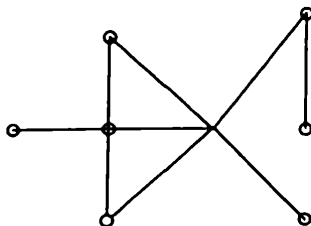
Time : Two Hours

Maximum : 60 Marks

Section A (Short Answer Type Questions)

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Find number of edges of $K_{m,n}$.
2. Draw the graph $K_6 - \{v\}$ where v is any vertex in K_6 .
3. Draw a 4-regular graph with ten vertices.
4. Define intersection of two graphs.
5. Let G be a simple graph in which there is no pair of adjacent edges. What can you say about the degree of the vertices in G ? Justify.
6. Draw the graph $K_{2,3,3}$.
7. Define eccentricity and radius.
8. Draw Peterson graph and find a trail of length 5.
9. When can you say that the complete graph $K_n, n \geq 3$ is Euler? Justify.
10. Prove that any subgraph of a planar graph is planar.
11. Find $K(G)$ for the graph.



12. How many different Hamiltonian cycles does K_n have?

(8 × 3 = 24 marks)

Turn over

Section B (Paragraph/Problem Type Questions)

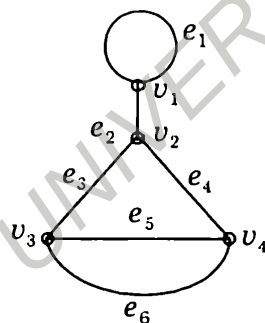
Answer at least **five** questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. If G is a simple planar graph then prove that G has a vertex v of degree less than 6.
14. Prove that the complete bipartite graph $k_{3,3}$ is non-planar.
15. Let G be a graph with n vertices, where $n \geq 2$. Then prove that G has atleast two vertices which are not cut vertices.
16. Explain the Konigsberg bridge problem.
17. Prove that G is connected if and only if it has a spanning tree.
18. Let G be a graph with n vertices. Then prove that G is a tree if and only if G is a connected graph with $n - 1$ edges.
19. Define (i) adjacency matrix of a graph G ; (ii) incidence matrix of a graph G . Find the adjacency and incidence matrix of the following graph G .



(5 × 5 = 25 marks)

Section C (Essay Type Questions)

Answer any **one** question.

The question carries 11 marks.

20. Prove the following :
 - (i) A connected graph G has an Euler tail if and only if it is atmost two odd vertices.
 - (ii) A simple graph G is Hamiltonian if and only if it closure $c(G)$ is Hamiltonian.
21. Prove that, a non-empty graph with atleast two vertices is bipartite if and only if it has no odd cycle.

(1 × 11 = 11 marks)